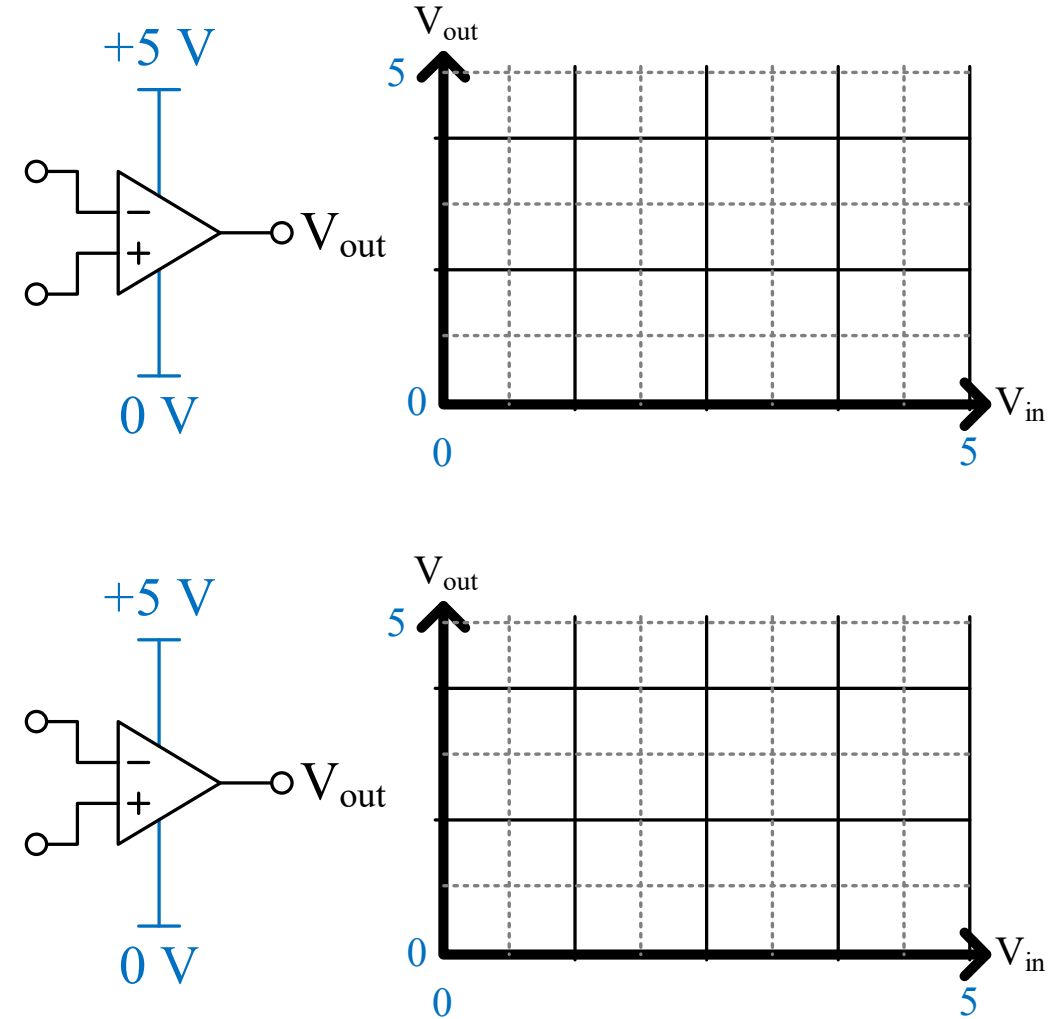


2.2 – Opamps: Comparators and Schmitt Triggers

Comparator

- Open-loop configuration
- Analog in $\rightarrow$ Digital out
- $V_{\text{out}} = \begin{cases} V_{\text{DD}}, V_+ > V_- \\ V_{\text{SS}}, V_+ < V_- \end{cases}$, i.e. it “hits the rails”
- Performs analog “greater than” function
- E.g.: A digital system with a light sensor
 - '0' = 0 V, '1' = 5 V
 - Light sensor output voltage exceeds 2 V during the day
 - Make 'daytime' and 'nighttime' detectors

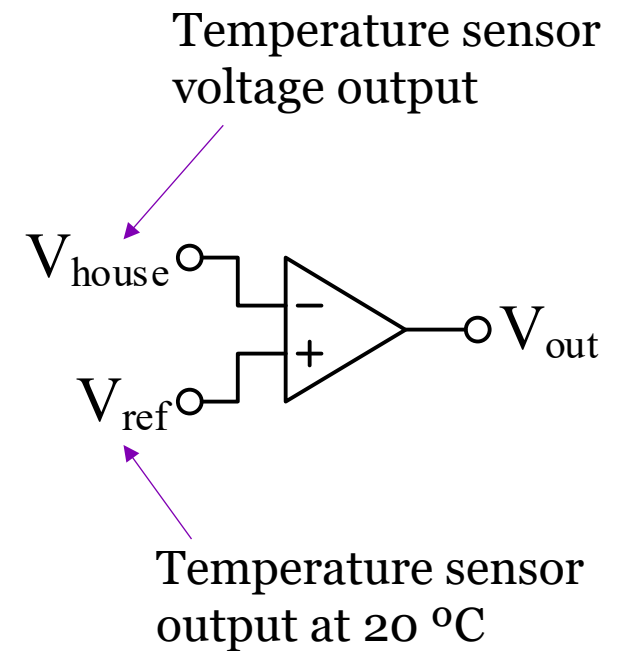
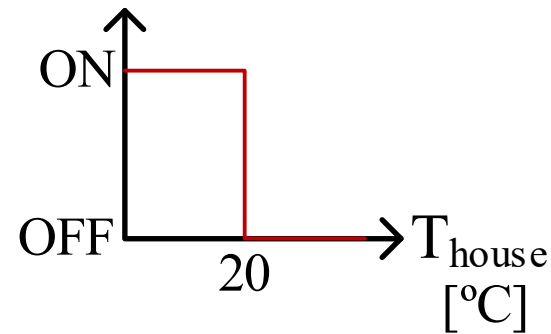
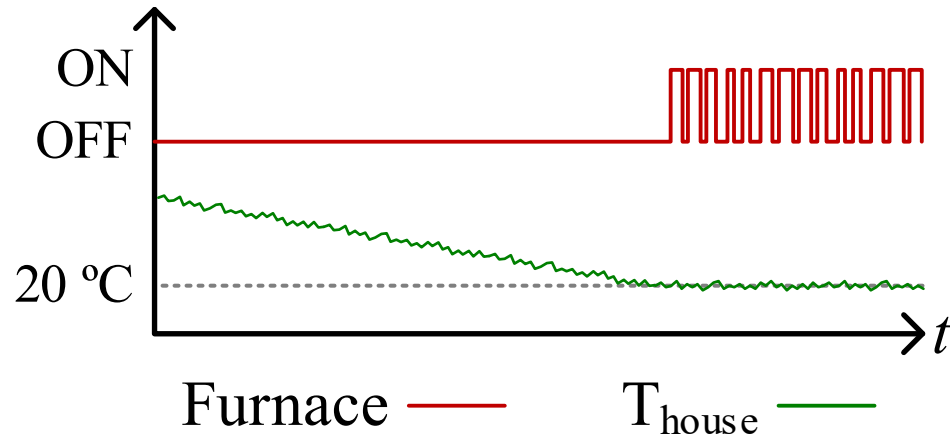


<https://everycircuit.com/circuit/5638772460617728>



Problem with Comparators

- Consider a house with a furnace and thermostat set to 20 °C
- Heating Strategy #1: Furnace ON when $T_{\text{house}} < 20\text{ °C}$

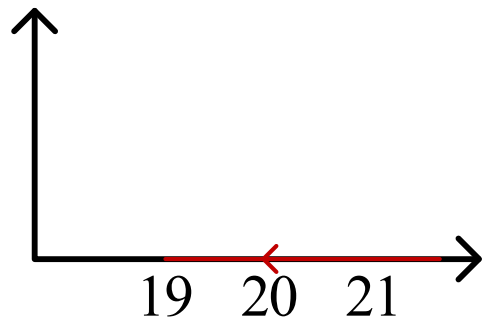
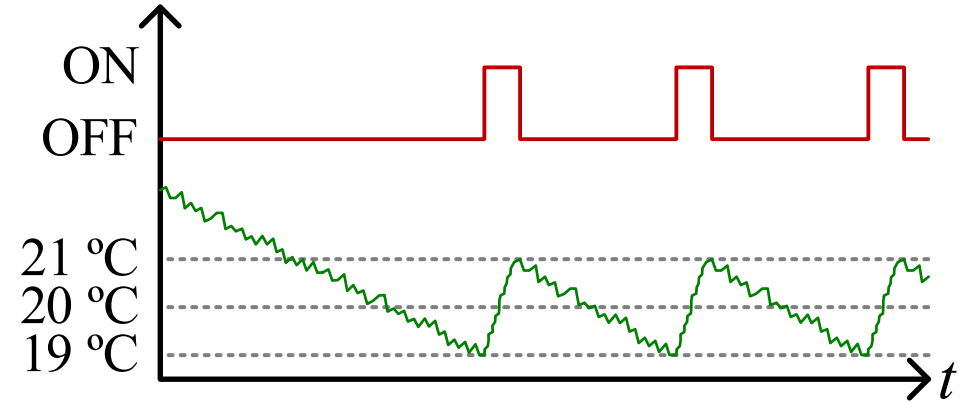


- Not a great strategy: Output exhibits “short cycling”, “output bounce”, “chatter”

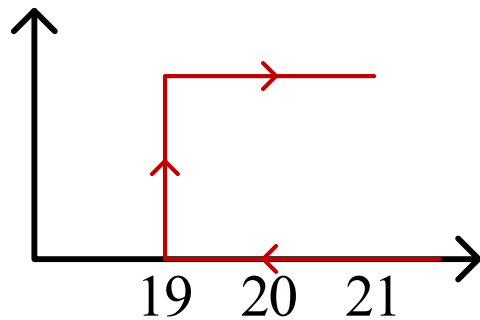


Hysteresis

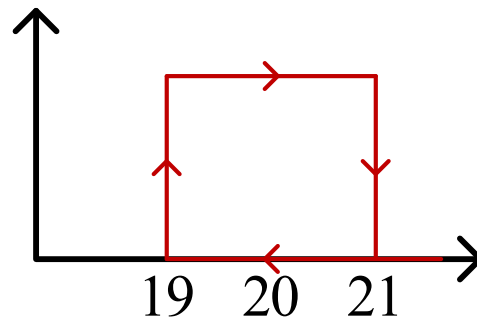
- Heating Strategy #2: Apply a range to slightly over-heat and over-cool
- Called **hysteresis**: Output depends on inputs *and* current state (output)



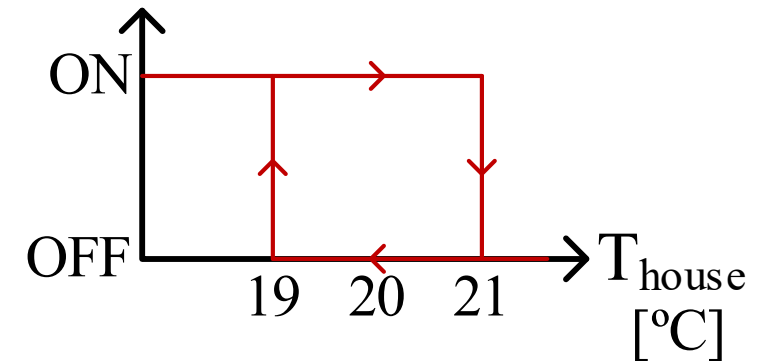
1) House cools to 19 °C



2) Furnace turns on and heats to 21 °C



3) Furnace turns off and cools to 19 °C

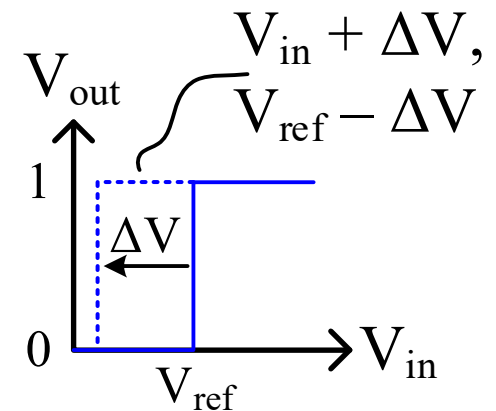
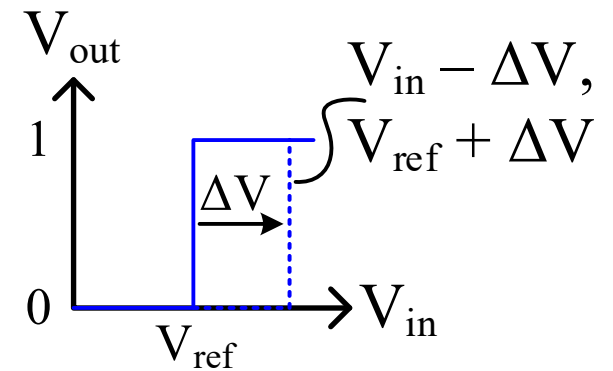
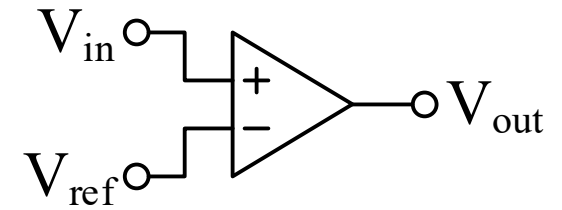


4) Complete heating cycle curve

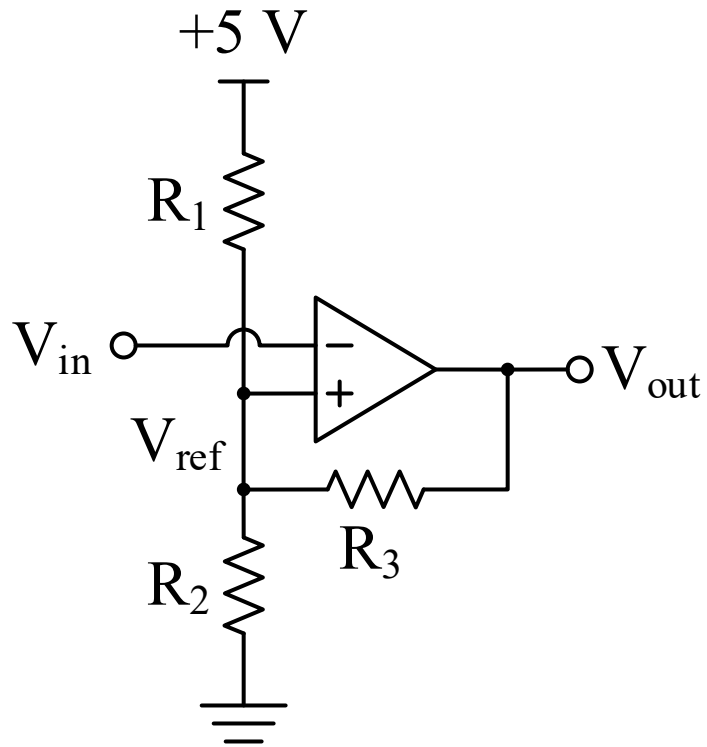


Adding Hysteresis to a Comparator

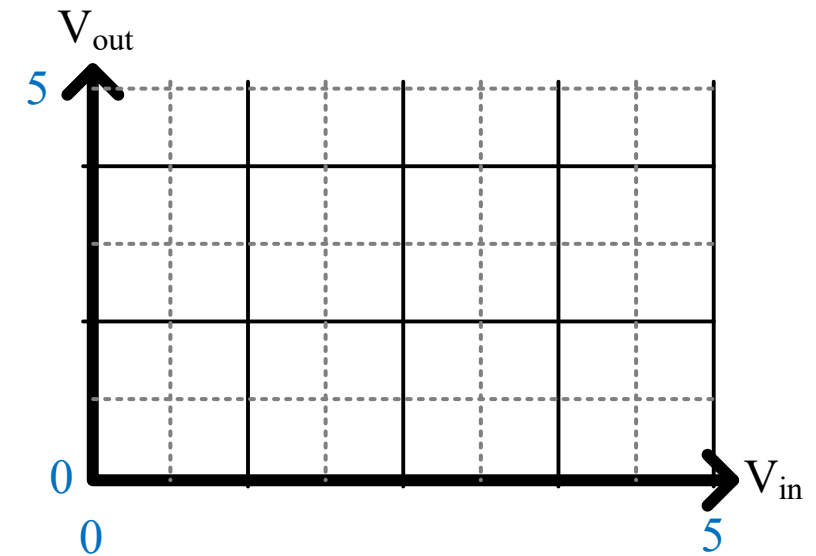
- Need to move transition point by $\pm\Delta V$ depending on current state
 - Can either shift V_{in} or V_{ref} :
- Recall that output always flips to '1' when $V_+ > V_-$
- Consider noninverting comparator
 - Move transition point right by ΔV , apply $V_{in} - \Delta V$ or $V_{ref} + \Delta V$
 - Move transition point left by ΔV , apply $V_{in} + \Delta V$ or $V_{ref} - \Delta V$
 - Choose whichever is most convenient to implement
- Same concept for inverting comparator, but opposite implementation



Inverting Schmitt Trigger

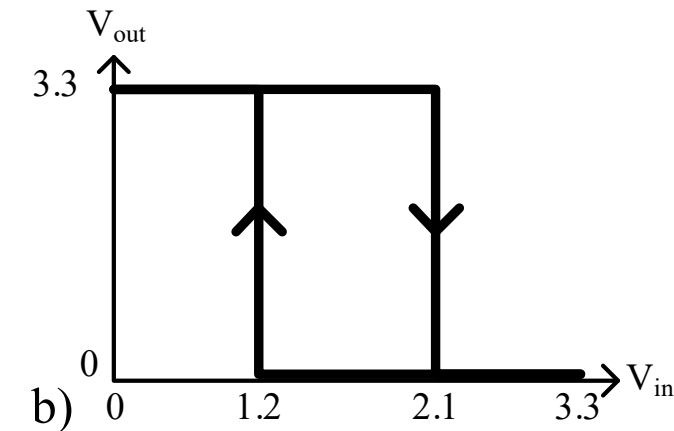
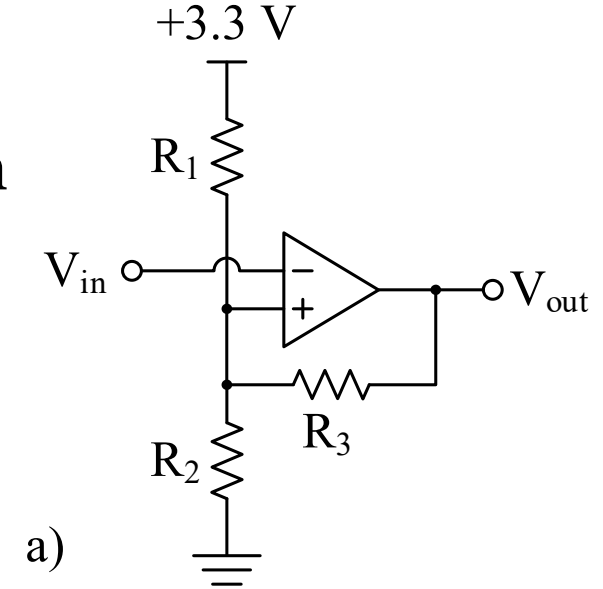


- A comparator with positive feedback
 - Makes the output 'stick' to one extreme or the other



Inverting Schmitt Trigger Example

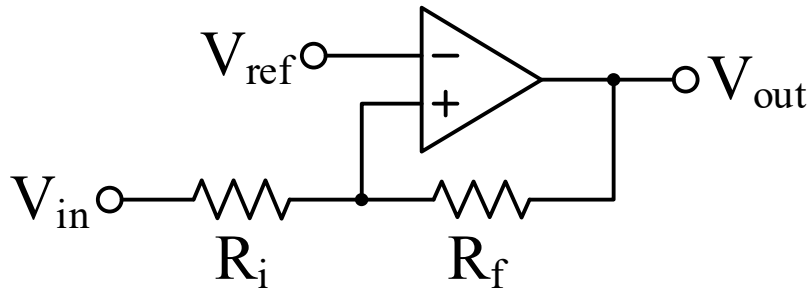
- Design the Schmitt trigger circuit in a) to achieve the transition points shown in b). The current through any resistor should not exceed 1 mA.



<https://everycircuit.com/circuit/6201722414039040>



Noninverting Schmitt Trigger



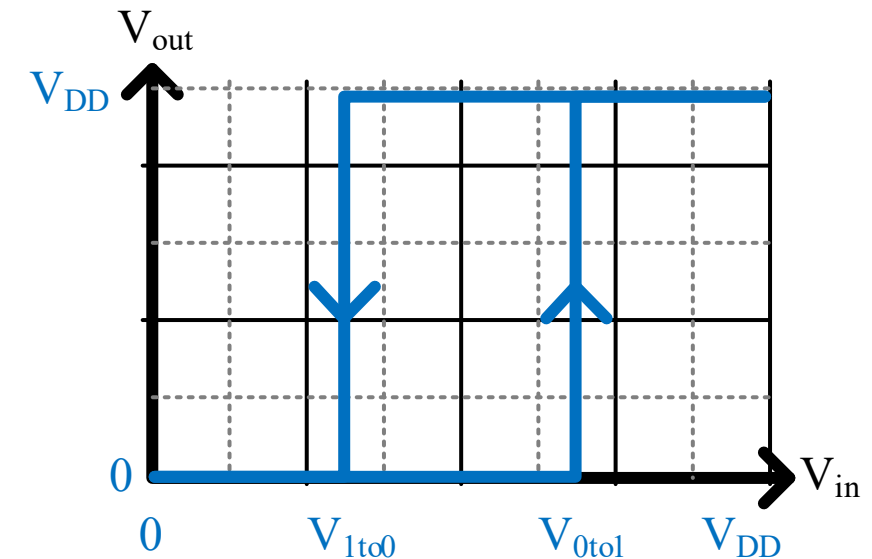
$$V_{1 \rightarrow 0} = \left(1 + \frac{R_i}{R_f}\right) V_{\text{ref}} - \left(\frac{R_i}{R_f}\right) V_{\text{DD}}$$

$$V_{0 \rightarrow 1} = \left(1 + \frac{R_i}{R_f}\right) V_{\text{ref}}$$

$$\frac{R_i}{R_f} = \frac{V_{0 \rightarrow 1} - V_{1 \rightarrow 0}}{V_{\text{DD}}}$$

$$V_{\text{ref}} = \frac{V_{0 \rightarrow 1}}{\left(\frac{V_{0 \rightarrow 1} - V_{1 \rightarrow 0}}{V_{\text{DD}}}\right) + 1} = \frac{V_{0 \rightarrow 1}}{\frac{R_i}{R_f} + 1}$$

- Output “drags” V_+ up or down (instead of V_{ref})
- Exercise: Derive transition points using basic opamp assumptions



LTSpice Modeling Challenge



- Create and tune inverting and noninverting Schmitt triggers to:
 - Act like a comparator with $V_{\text{ref}} = 2 \text{ V}$
 - Reject the noise on the given signal below using minimal hysteresis
 - Use $V_{\text{DD}} = 5 \text{ V}$ and $V_{\text{SS}} = 0 \text{ V}$
 - Minimum timestep = $10 \mu\text{s}$
- This SPICE directive creates a triangle wave from 1 V to 4 V with about $50 \text{ mV}_{\text{pk-pk}}$ of noise superimposed:
 - **Bsense SENSE 0 V = 1 + 3*(1-abs(2*time-2*floor(time+0.5))) + 50m*white(2*pi*20k*time)**
 - Press '.' to insert a SPICE directive and type in that line
 - Name whatever net you want this applied to as "SENSE"

